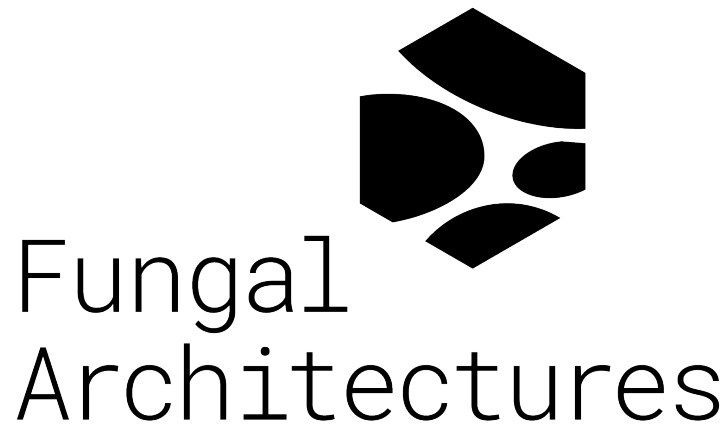


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Sensorial fusion and information processing primitives

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Description: In this deliverable we report about implementation of massive-parallel computing circuits with simulated and living mycelium networks. The report includes results on implementation of action-potential like spiking gates in a single fungal colony, simulation of electrical Boolean circuits, experimental laboratory mining of Boolean circuits in substrates colonised by fungi and experimental laboratory implementation of fungal circuits in responsive insoles.

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1 BACKGROUND

A vibrant field of unconventional computing aims to employ space-time dynamics of physical, chemical and biological media to design novel computational techniques, architectures and working prototypes of embedded computing substrates and devices. Interaction-based computing devices, is one of the most diverse and promising families of the unconventional computing structures. They are based on interactions of fluid streams, signals propagating along conductors or excitation wave-fronts. Typically, logical gates and their cascade implemented in an excitable medium are ‘handcrafted’ to address exact timing and type of interactions between colliding wave-fronts. The artificial design of logical circuits might be suitable when chemical media or functional materials are used. However, the approach might be not feasible when embedding computation in living systems, where the architecture of conductive pathways may be difficult to alter or control. In such situations an opportunistic approach to outsourcing computation can be adopted. The system is perturbed via two or more input loci and its dynamics if recorded at one or more output loci. A wave-front appearing at one of the output loci is interpreted as logical truth or ‘1’. Thus the system with relatively unknown structure implements a mapping $\{0, 1\}^n \rightarrow \{0, 1\}^m$, where n is a number of input loci and m is a number of output loci, $n, m > 0$. The approach belong to same family of computation outsourcing techniques as *in materio* computing. Fungal colonies are characterised by rich typology of mycelium networks in some cases affin to fractal structures. Rich morphological features might imply rich computational abilities and thus worse to analyse from realising Boolean functions point of view. In numerical experiments we study implementation of logical gates via interaction of numerous travelling excitation waves, seen as as action potentials, on an image of a real fungal colony. Detailed results related to the Deliverable are presented in [roberts_mining_2021, 1, 2, 3].

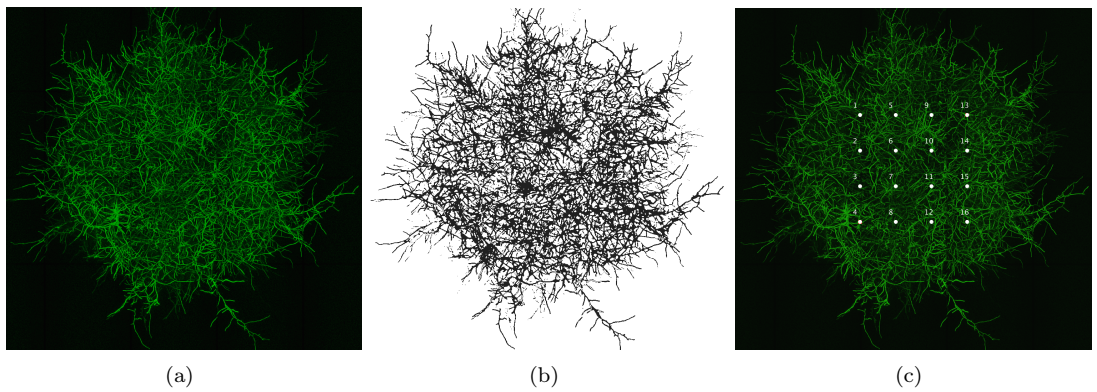


Figure 1: Image of the fungal colony, 1000×960 pixels used as a template conductive for FHN. (a) Original image, mycelium is seen as green pixels. (b) Conductive matrix C , conductive pixels are black. (c) Configuration of electrodes.

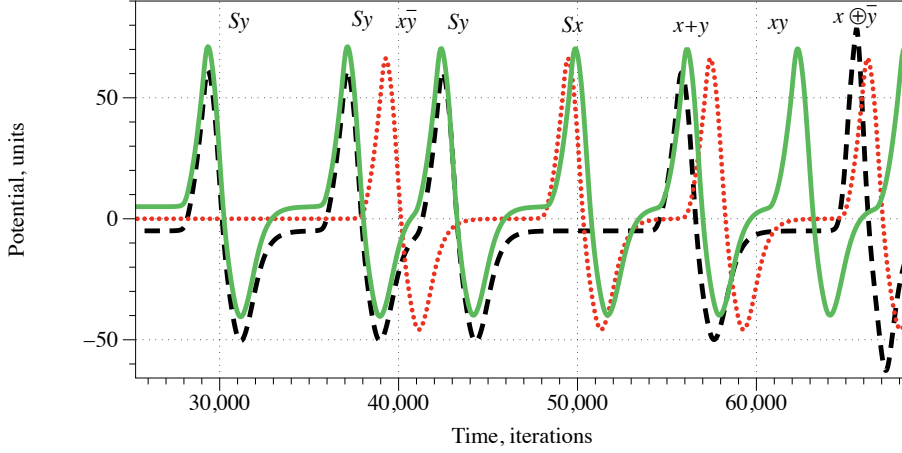


Figure 2: Fragment of electrical potential record on electrode 7 in response to inputs (01), black dashed line, (10), red dotted line, (11), solid green line, entered as impulses via electrodes $E_x = 5$ and $E_y = 15$. See locations of electrodes in Fig. 1d. To make the individual plots visible in places of exact overlapping, we added potential -5 to recording in response to input (01) and and potential 5 to recording in response to input (11).

2 SPIKES BASED BOOLEAN GATES IN A MODEL OF A SINGLE FUNGAL COLONY

A fungal colony maintains its integrity via flow of cytoplasm along mycelium network. This flow, together with possible coordination of mycelium tips propagation, is controlled by calcium waves and associated waves of electrical potential changes. We propose that these excitation waves can be employed to implement a computation in the mycelium networks. We use FitzHugh-Nagumo model to imitate propagation of excitation in a single colony of *Aspergillus niger* (Fig. 1ab). The waves of excitation are recorded by an array of simulated electrodes (Fig. 1c).

Boolean values are encoded by spikes of extracellular potential (Fig. 2). We represent binary inputs by electrical impulses on a pair of selected electrodes and we record responses of the colony from sixteen electrodes. We derive sets of two-inputs-on-output logical gates implementable the fungal colony and analyse distributions of the gates.

In paper [1] we have demonstrated how sets of logical gates can be implemented in single colony mycelium networks via initiation of electrical impulses. The impulses travel in the network, interact with each other (annihilate, reflect, change their phase). Thus for different combinations of input impulses and record different combinations of output impulses, which in some cases can be interpreted as representing two-inputs-one-output functions.

To estimate a speed of computation we refer to Olsson and Hansson's [4] original study, in which they proposed that electrical activity in fungi could be used for communication with message propagation speed 0.5 mm/sec . Diameter of the colony (Fig. 1a), which experimental laboratory images has been used to run FHN model, is c. 1.7 mm . Thus, it takes the excitation waves initiated at a boundary of the colony up to 3-4 sec to span the whole mycelium network (this time is equivalent to c. 70K iterations of the numerical integration model). In 3-4 sec the mycelium network can compute up to a hundred logical gates. This gives us the rate of a gate per 0.03 sec , or, in terms of frequency this will be c. 30 Hz . The mycelium network computing can

not compete with existing silicon architecture however its application domain can be a unique of living biosensors (a distribution of gates realised might be affected by environmental conditions) and computation embedded into structural elements where fungal materials are used.

3 ELECTRICAL ANALOG BOOLEAN GATES

In numerical modelling and experimental laboratory setup we exploited principles of electrical analog computing [2]. TRUE and FALSE values are represented by above threshold and below threshold voltages. Due to the non-linearity of the conductive substrate along electrical current pathways between input and output electrodes, the input voltages are transformed and thus logical mappings are implemented. Detailed descriptions of these techniques can be found in [2].

The z -stacks of a single colony of *Aspergillus niger* fungus strain AR9#2 were converted to a 3D graph (Fig. 3). We modelled the colony as a resistive and capacitive (RC) network. RC networks are circuits consisting of resistances and capacitors, the most fundamental passive circuit elements needed to design from a low-pass filter up to an equivalent network of a nerve cell. The 3D graph was converted to the RC network, whose magnitudes are a function of the length of the connections.

Resistances were in the order of kOhms and capacitance in the order of pF. The positive voltage and ground nodes were randomly assigned from the sample and 1000 networks were created in each arrangement for analysis. SPICE analysis consisted of transient analysis using a two voltage pulses of 60 mV on the randomly assigned positive nodes. We modelled the fungal colony in serial RC networks and parallel RC networks. The output voltages were binarised with the threshold θ : $V > \theta$ symbolises logical TRUE otherwise FALSE.

There are 16 possible logical gates realisable for two inputs and one output. The gates implying input 0 and evoking a response 1, i.e. $f(0,0) = 1$, are not realisable because the simulated fungal circuit is passive. The remaining 8 gates are AND, OR, AND-NOT (x AND NOT y and NOT x AND y), SELECT (SELECT x and SELECT y) and XOR. In the model of serial RC networks, we found gates AND, SELECT and AND-NOT; no OR gates have been found. The number n of the gates discovered decreases by a power law with increase of θ . The frequency of AND gates oscillates, as shown in the zoom insert in Fig. 4a, most likely due to its insignificant presence in the samples. The oscillations reach near zero base when θ exceeds 0.001. In the model of parallel RC networks we only found the gates AND, SELECT and OR. The number of OR gates decreases quadratically and becomes nil when $\theta > 0.03$. The number of AND gates increases near linearly with increase of θ . The number of SELECT gates reaches its maximum at $\theta = 0.023$, and then starts to decrease with the further increase of θ . To conclude, mycelium bound composites can act as computing media and implement a wide range of Boolean circuits, thus opening a new perspective in biological analog and hybrid computing.

4 EXPERIMENTAL MINING OF BOOLEAN CIRCUITS

Living substrates are capable for nontrivial mappings of electrical signals due to the substrate nonlinear electrical characteristics. This property can be used to realise Boolean functions. Input logical values are represented by amplitude or frequency of electrical stimuli. Output logical values are decoded from electrical responses of living substrates. We demonstrate how logical circuits can be implemented in mycelium bound composites. The mycelium bound composites (fungal materials) are getting growing recognition as building, packaging, decoration and clothing materials. Presently the fungal materials are passive. To make the fungal materials adaptive,

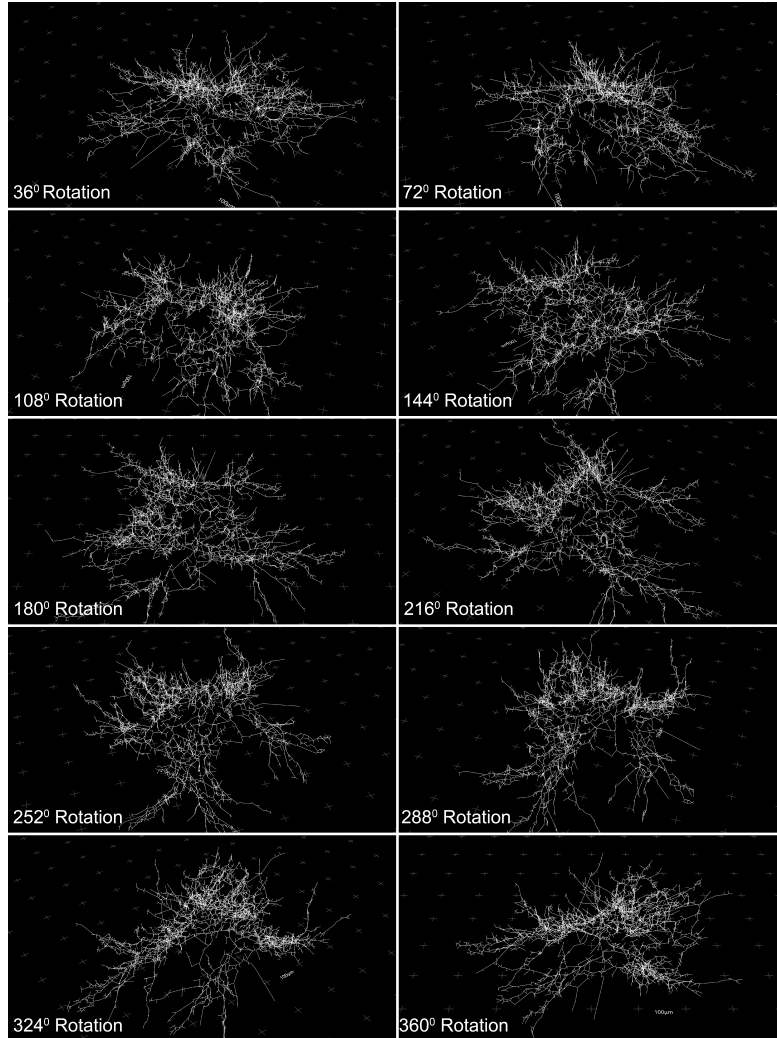


Figure 3: Perspective views of the 3D Graph. Each frame shows the graph after a 36° rotation around the z -axis with origin located approximately in the centre of the colony, on the $x - y$ plane indicated with registration marks.

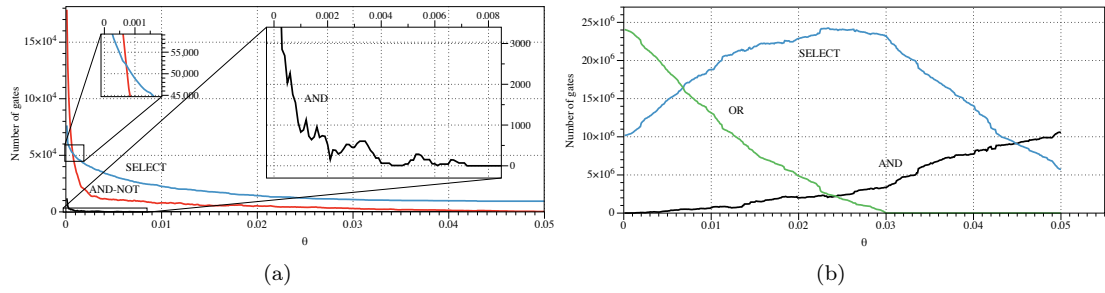


Figure 4: Occurrences of the gates from the groups AND, black, OR, green, AND-NOT, red, and SELECT, blue, for $\theta \in [0.0001, 0.05]$, with θ increment 0.0001, in (a) fungal colony modelled with serial RC networks, (b) fungal colony modelled with parallel RC networks.

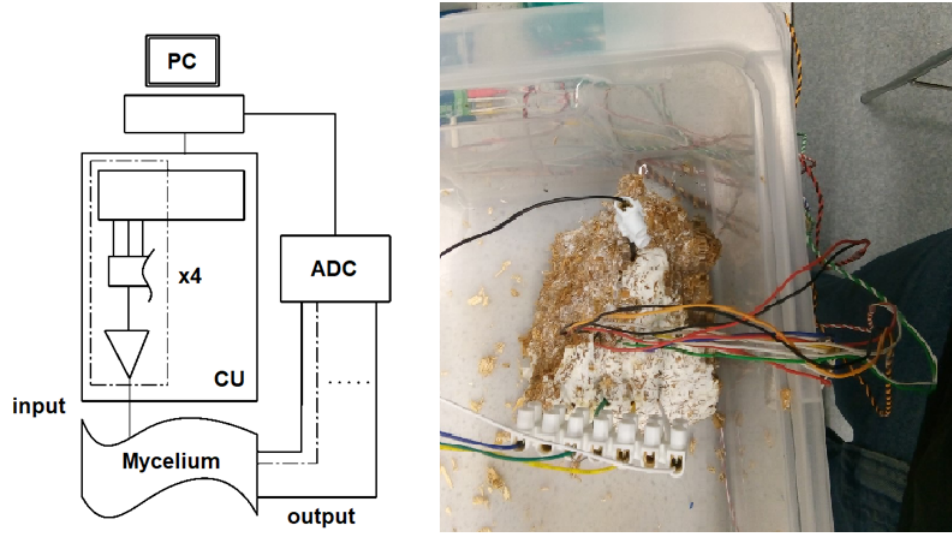


Figure 5: Left: Schematic of the mycelium communications system; PC — laptop for generating sequences; CU – control unit, dashed section is a breakdown of a single channel; ADC — analogue to digital converter. Right: experimental set up.

i.e. sensing and computing, we should embed logical circuits into them. We demonstrate experimental laboratory prototypes of many-input Boolean functions implemented in fungal materials from oyster fungi *P. ostreatus*. We characterise complexity of the functions discovered via complexity of the space-time configurations of one-dimensional cellular automata governed by the functions. We show that the mycelium bound composites can implement representative functions from all classes of cellular automata complexity including the computationally universal. The results presented will make an impact in the field of unconventional computing, experimental demonstration of purposeful computing with fungi, and in the field of intelligent materials, as the prototypes of computing mycelium bound composites.

A hemp shavings substrate was colonised by the mycelium of the grey oyster fungi, *P. ostreatus* (Ann Miller's Speciality Mushrooms Ltd, UK). Recordings were carried out in a stable indoor

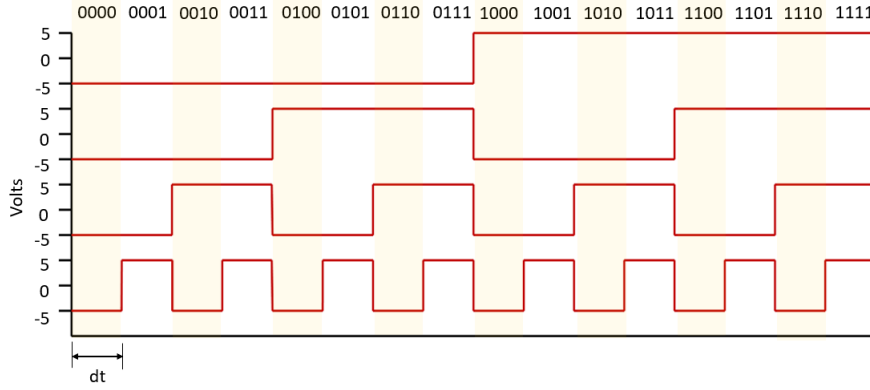


Figure 6: Timing diagram and associated Boolean strings for four inputs into the mycelium substrate, time step is one hour.

environment with the temperature remaining stable at $22 \pm 0.5^\circ$ and relative humidity of air $40 \pm 5\%$. The humidity of the substrate colonised by fungi was kept at c. 70-80%.

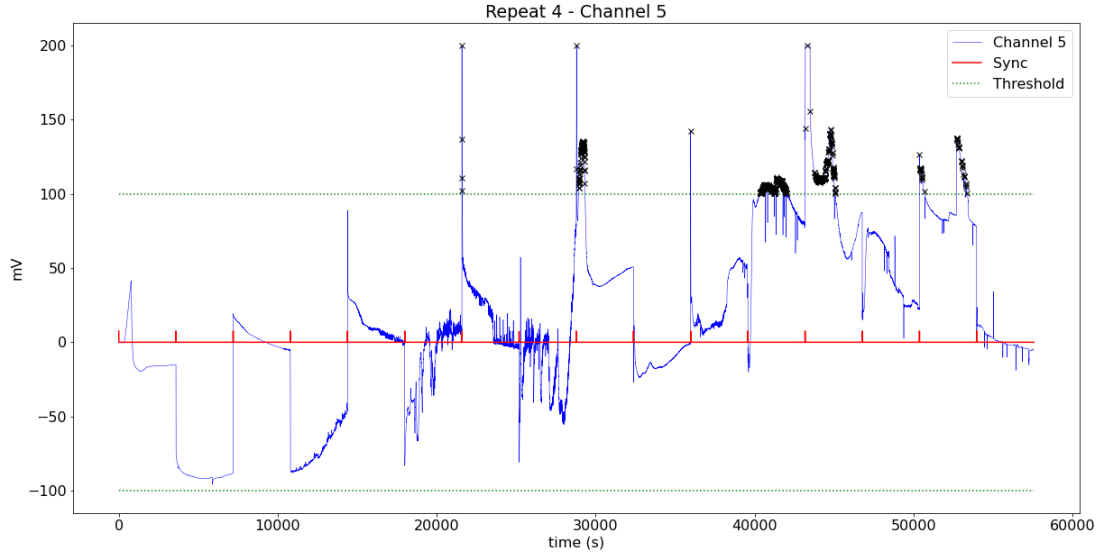
Hardware was developed that was capable of sending sequences of 4 bit strings to a mycelium substrate. The strings were encoded as step voltage inputs where -5 V denoted a logical 0 and 5 V a logical 1. The hardware was based around an Arduino Mega 2560 (Elegoo, China) and a series of programmable signal generators, AD9833 (Analog, USA). The 4 input electrodes were 1 mm diameter platinum rods inserted to a depth of 50 mm in the substrate in a straight line with a separation of 20 mm. Data acquisition (DAQ) probes were placed in a parallel line 50 mm away separated by 10 mm. The electron sink and source was placed 50 mm on from DAQ probes. There were 7 DAQ differential inputs from the mycelium substrate to a Pico 24 (Pico Technology, UK) analogue-to-digital converter (ADC), the 8th channel was used to pass a pulse to the ADC on every input state change, see Fig. 5 for a schematic of the apparatus. The substrate and probes were placed in a semi-sealed container. After each experimental repeat the substrate was sprayed with water, left for an hour and then the next repeat was conducted. There were a total of 14 repeats.

A sequence of 4 bit strings counting up from binary 0000 to 1111, with a state change every hour, were passed into the substrate, see Fig. 6 for timing details. In all 14 repeats of the experiment were done on the same substrate to capture changes in structure of the growing mycelium. Samples from 7 channels were taken at 1 Hz over the whole duration of a given experimental run. Peaks for each channel were located for a set of 32 thresholds, from 20 mV to 175 mV with step 5 mV, for each input state, 0000 to 1111.

Boolean strings were extracted from the data, where a logic '1' was noted for a channel if it had a peak outside the threshold band for a particular state else, a value of '0' was recorded, the polarity of the peak was not considered.

The strings for each experimental repeat were stored in their respective Boolean table. To extract state graphs, a state/node was defined as the string of output values from each channel at each input state, transitions/edges were defined as a change in input state. This led to a total of 448 state graphs. The sum of products (SOP) Boolean functions were calculated for each output channel. For each repeat there were 7 channels and 32 thresholds giving total of 3136 individual truth tables.

See Fig. 7 for SOP extraction. If a peak is found in Fig. 7a during an input state then this is considered a logical 1, highlighted in yellow in table Fig. 7b are the thresholded values for



(a)

Inputs	Output channels						
0000	0	0	0	0	0	0	0
0001	0	1	1	0	1	0	0
0010	0	1	1	0	1	0	0
0011	0	1	1	0	1	0	1
0100	0	0	1	0	1	0	0
0101	1	1	1	0	1	0	1
0110	1	0	1	0	1	0	1
0111	0	1	1	0	1	0	1
1000	1	1	1	0	1	0	1
1001	1	1	0	0	1	0	1
1010	0	0	1	0	0	0	0
1011	1	1	1	1	1	0	1
1100	1	1	1	0	0	1	1
1101	1	0	1	0	0	0	1
1110	1	0	1	0	1	0	1
1111	0	0	0	0	0	0	0



$$A\bar{C} + A\bar{B}D + B\bar{C}\bar{D} + B\bar{C}D$$

(b)

Figure 7: Workflow example. (a) The measurements taken by channel 5 of the DAQ in blue, the synchronisation signal is shown red which marks the state change, threshold band shown in green, peaks outside this band are highlighted with ‘x’ marker. (b) The truth and the function extracted.

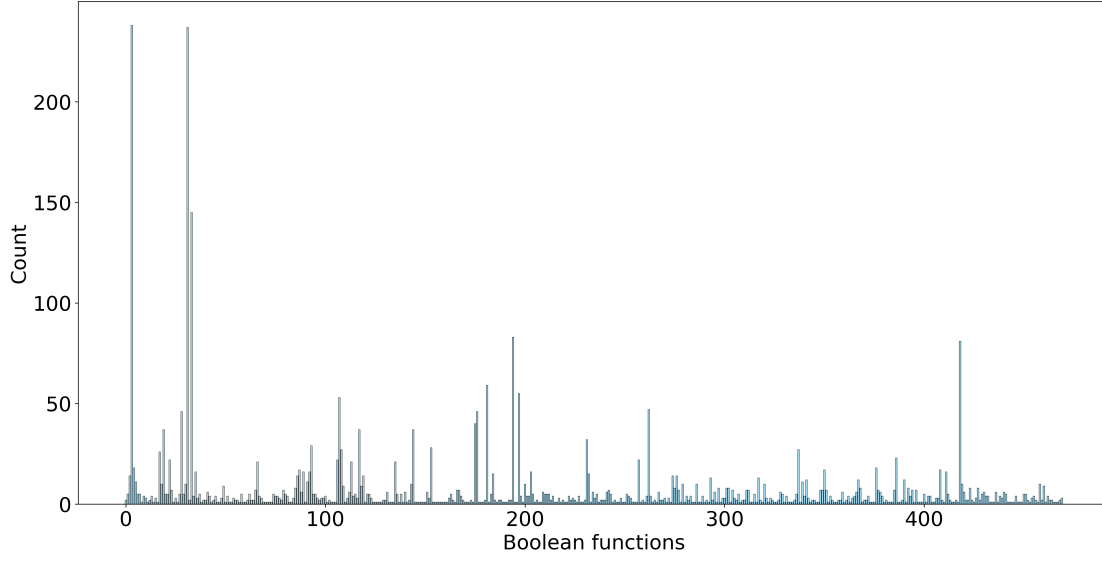


Figure 8: Counts of realised Boolean functions discovered in laboratory experiments. Horizontal axis is a decimal representation of functions. Vertical axis is a number of functions discovered in experiments.

Table 1: Top 16 highest occurring Boolean functions.

Count		Boolean function
145	F_1	$\overline{A} + \overline{B} + \overline{C} + \overline{D}$ (NAND)
83	F_2	$\overline{A}\overline{B} + \overline{A}\overline{C} + \overline{A}\overline{D} + \overline{A}B + \overline{A}C + \overline{A}D + \overline{B}\overline{C} + \overline{B}\overline{D} + \overline{B}C + \overline{B}D + \overline{C}\overline{D} + \overline{C}D + \overline{D}\overline{C} + \overline{D}D$
81	F_3	$\overline{A}\overline{C}\overline{D} + \overline{A}B\overline{C} + \overline{A}B\overline{C} + \overline{A}B\overline{D}$
59	F_4	$\overline{A}\overline{C} + \overline{A}\overline{D} + \overline{A}C + \overline{C}\overline{D} + \overline{A}D + \overline{B}D + \overline{C}D$
55	F_5	$\overline{A}B + \overline{C}\overline{D} + \overline{A}D$
53	F_6	$\overline{A}\overline{B}\overline{C}D$
47	F_7	$\overline{B}\overline{D} + \overline{C}\overline{D} + \overline{A}D + \overline{B}C\overline{D}$
46	F_8	$\overline{A}B\overline{C}\overline{D}$
46	F_9	$A + B + C + D$ (OR)
40	F_{10}	$\overline{A}\overline{B} + \overline{A}\overline{D} + \overline{A}B + \overline{B}\overline{D} + \overline{A}D + \overline{B}D + \overline{C}D$
37	F_{11}	$\overline{A}\overline{B}\overline{C}\overline{D}$
37	F_{12}	$\overline{A}\overline{D} + \overline{A}B + \overline{B}\overline{C} + \overline{A}D + \overline{B}C\overline{D}$
37	F_{13}	$\overline{A}\overline{B} + \overline{A}\overline{C} + \overline{A}\overline{D} + \overline{A}D + \overline{B}\overline{D} + \overline{C}D\overline{A}B\overline{C} + \overline{B}C\overline{D}$
32	F_{14}	$\overline{A}\overline{D} + \overline{A}B + \overline{B}\overline{D} + \overline{A}C + \overline{C}\overline{D} + \overline{A}D + \overline{A}\overline{B}\overline{C} + \overline{B}C\overline{D}$
29	F_{15}	$\overline{C} + \overline{A}\overline{B} + \overline{A}\overline{D} + \overline{A}B + \overline{B}D\overline{A}D + \overline{B}D$
28	F_{16}	$\overline{A}B + \overline{A}\overline{C} + \overline{B}D + \overline{B}C\overline{D} + \overline{A}\overline{B}\overline{C}$

channel 5, the resulting truth table is then reduced to a sum products shown below the table.

We have discovered total of 3136 4-inputs-1-output Boolean functions. 470 unique functions are presented in Supplementary Materials. Figure 8 shows the Boolean function distribution. The two peak values were logical FALSE, $n = 238$, and logical TRUE, $n = 237$. The highest occurring non-trivial gate was $\overline{A} + \overline{B} + \overline{C} + \overline{D}$, $n = 145$. The top 16 occurring non-trivial

Boolean functions are listed in table 1. The only single gate functions found were for NAND ($\overline{A+B+C+D}$), $n = 145$, OR ($A+B+C+D$), $n = 46$, and AND ($ABCD$), $n = 8$. More details are provided in [5].

5 EXEMPLAR APPLICATION IN FUNGAL INSOLES

Whilst living fungal Boolean circuits could be used in computing devices in future, it would be useful to test have some application domains at present. Here we represent results of experiments on living fungal insoles, further detail can be found here [3].

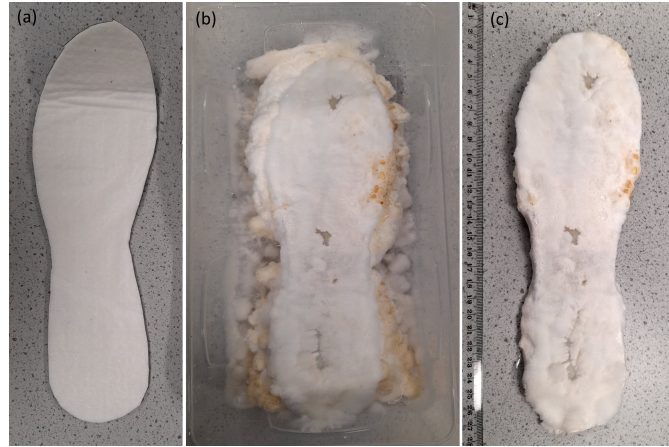


Figure 9: (a) capillary matting cut into insole pattern (b) insole on bed of spawn (c) well colonised insole

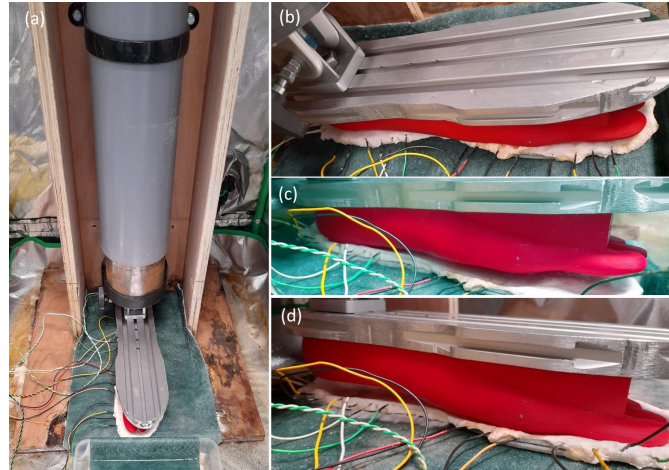


Figure 10: Bespoke insole test rig (a) setup inside growth tent (b) weight uniformly distributed via pivot joint on prosthetic foot (c) heel bias (d) toes bias.

Mycelium bound composites are promising materials for a diverse range of applications including wearables and building elements. Their functionality surpasses some of the capabilities

of traditionally passive materials, such as synthetic fibres, reconstituted cellulose fibres and natural fibres. Thereby, creating novel propositions including augmented functionality (sensory) and aesthetic (personal fashion). Biomaterials can offer multiple modal sensing capability such as mechanical loading (compressive and tensile) and moisture content. To assess the sensing potential of fungal insoles we undertook laboratory experiments on electrical response of bespoke insoles made from capillary matting colonised with oyster fungi *Pleurotus ostreatus* (Fig. 9) to compressive stress which mimics human loading when standing and walking (Fig. 10). We have shown changes in electrical activity with compressive loading. The results advance the development of intelligent sensing insoles which are a building block towards more generic reactive fungal wearables. Using FitzHugh-Nagumo model we numerically illustrated how excitation wave-fronts behave in a mycelium network colonising an insole and shown that it may be possible to discern pressure points from the mycelium electrical activity.

Electrical activity (spiking) was recorded in mycelium bound composites fabricated into insoles. The number and periodicity of electrical spikes change when the mycelium is subjected to compressive loading. We have shown that it might be possible to discern the loading from the electrical response of the fungi to stimuli [3]. The results advance the development of intelligent sensing insoles which are a building block towards more generic reactive fungal wearables. Electrical activity changes in both spatial and temporal domains. Using FitzHugh-Nagumo model we numerically illustrated how excitation wave-fronts behave in a mycelium network colonising an insole and shown that it might be possible to discern pressure points from the mycelium electrical activity. Fungal based insoles offer augmented functionality (sensory) and aesthetic (personal fashion). We presented results of scoping experiments on living biowearables. The results open new horizons in exploring feasibility of living fungal materials in everyday life. Directions of future research will involve bench marking of the prototypes and testing them in real life.

6 CONCLUSION

In a series of computer models and laboratory experiments we have demonstrated that mycelium networks are capable for sensorial fusion, information processing and realisation of logical circuits. The findings open new exciting pathways for integration of sensing and computing fungal devices into fungal materials, buildings and architectures.

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